EPFL

Group 2.A: Surveyor Drone

Charlotte Heibig
Ariane Priou

EPFL - 30.01.2024

Overview

Devices

- Phone
- Drone
- Gamepad (optional)

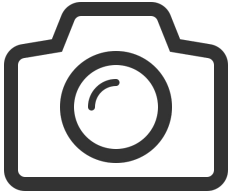
Emergency / safety features

Application

Autonomous or manual flight mode

Survey

Download data on firebase



Takes pictures and videos live video feed

Workload's distribution

A.Priou

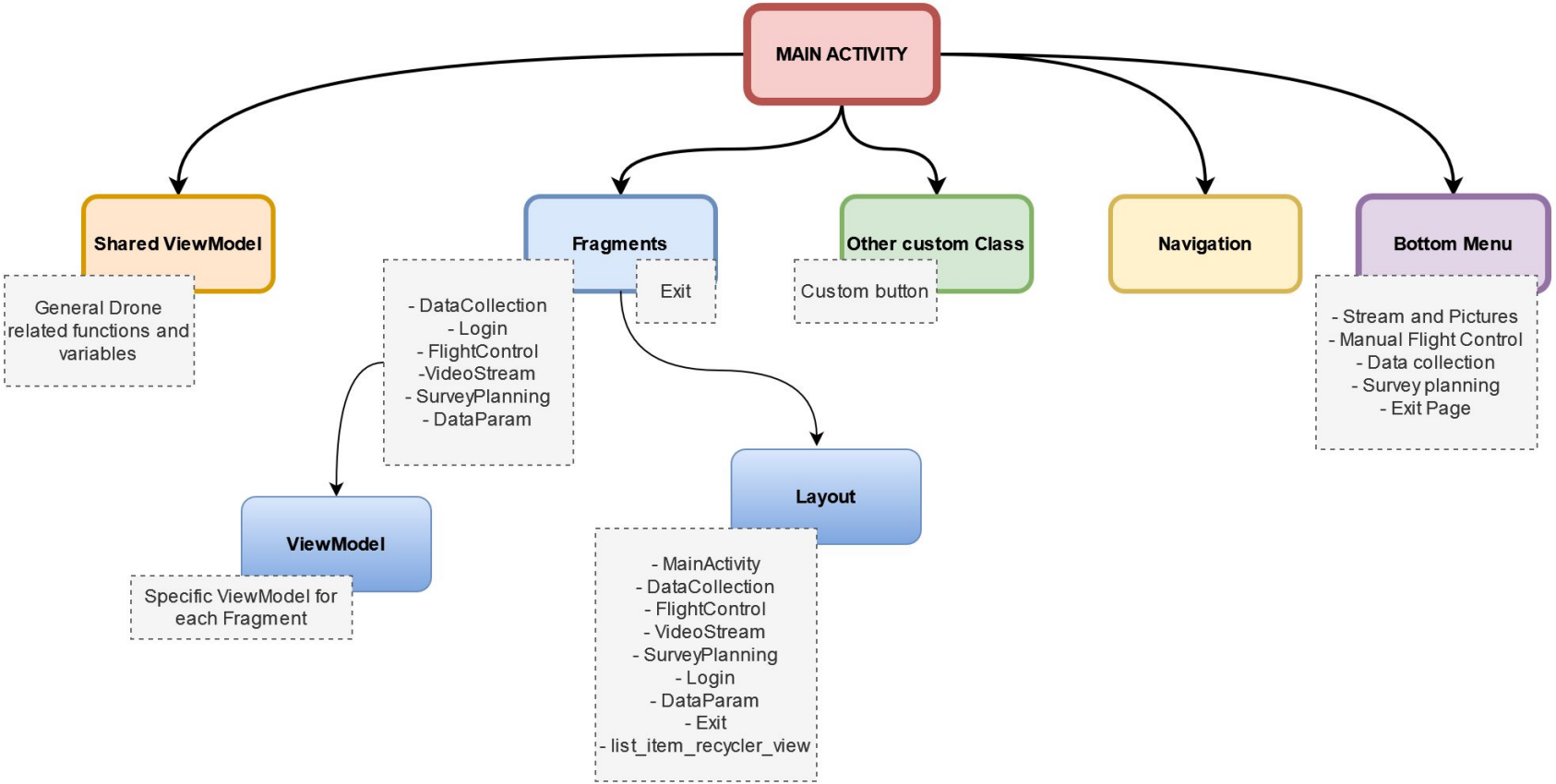
- UI
- Navigation
- Photos & videos
- Exit page
- Emergency / safety features

C. Heibig

- Login and firebase
- Stream
- upload of the map
- waypoint placement
- Autonomous flight control
- Manual flightControl

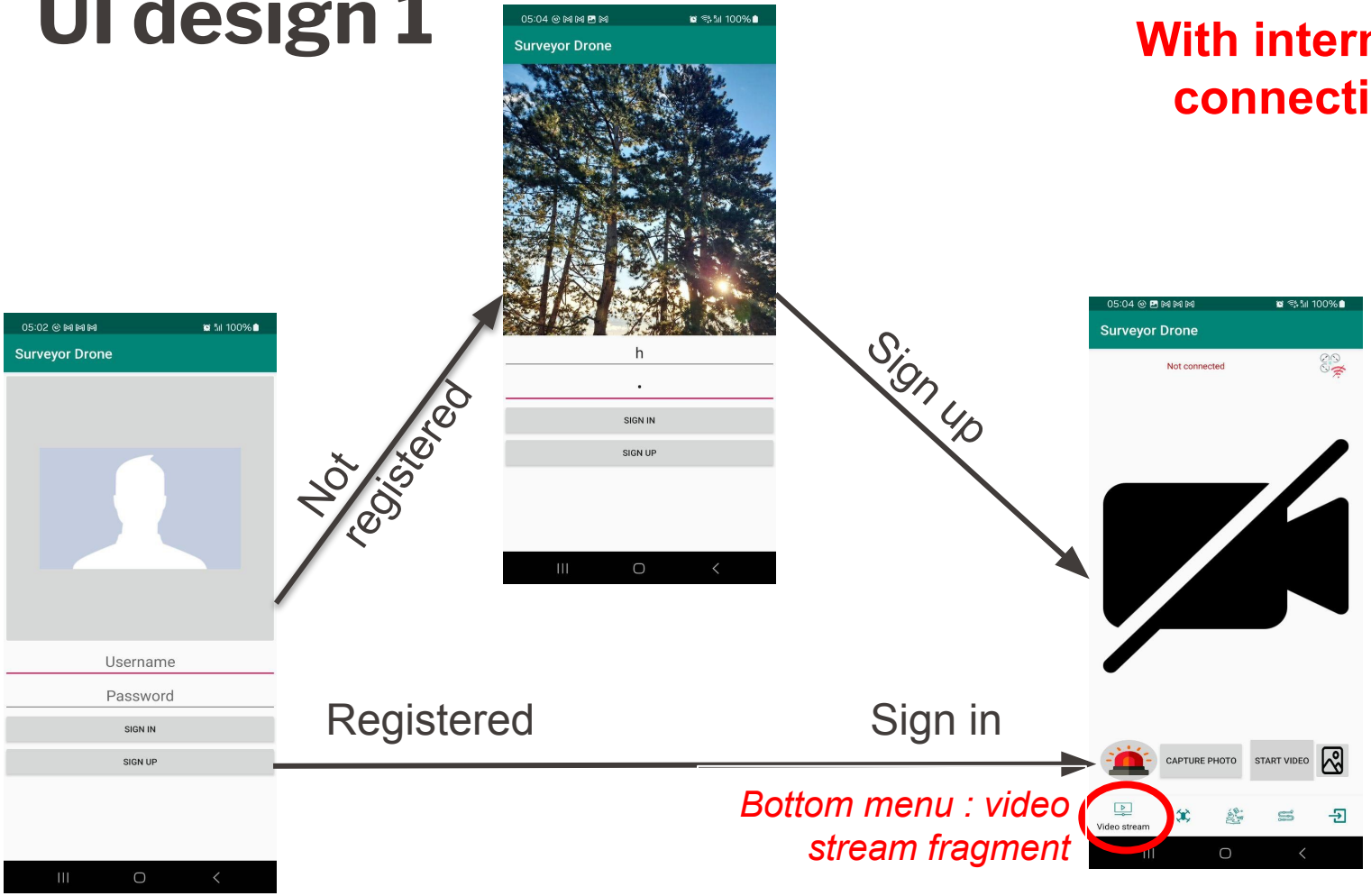
- Sadly the third member has left the course

Architecture and key features



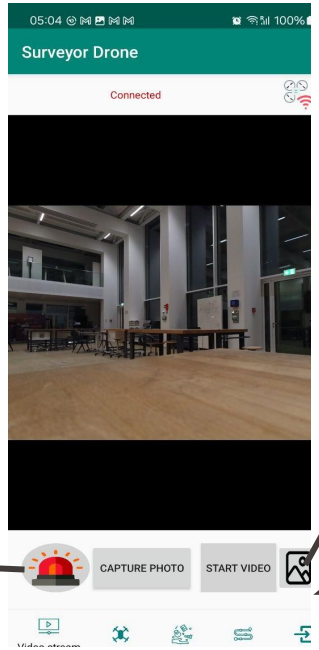
UI design 1

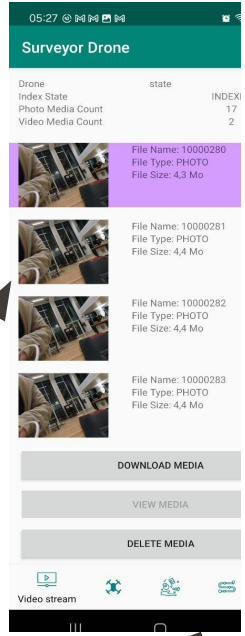
With internet connection

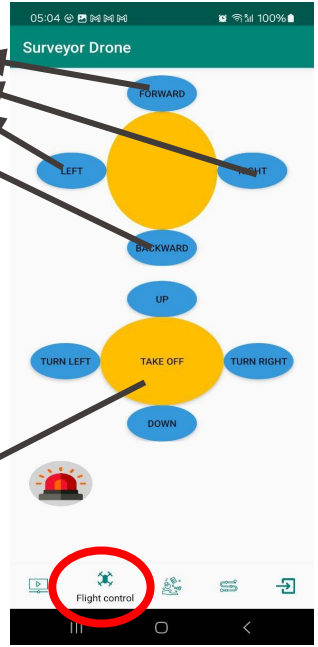


UI design 2

Connection to the drone







Emergency button

Manage photos/videos

Use back button of the phone

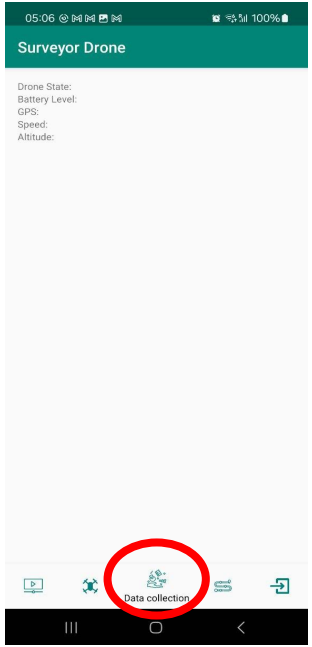
Control manually the drone

Take off/ Land

Bottom menu : flight control fragment

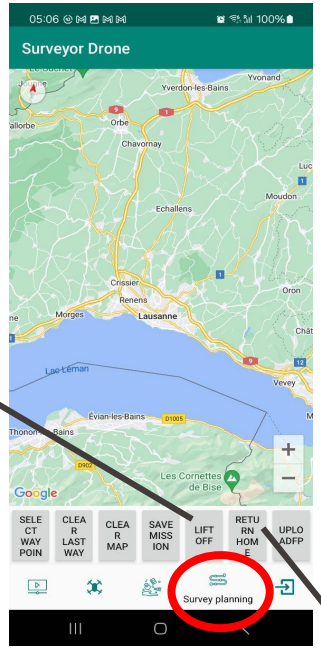
UI design 2

Connection to the drone



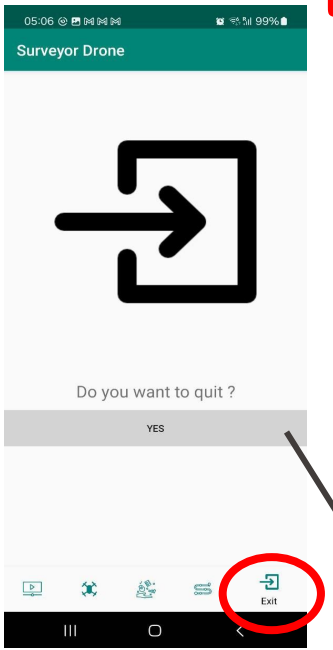
Bottom menu : data collection fragment

Begin mission (when drone is flying)



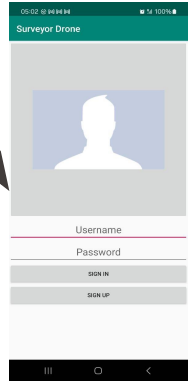
Bottom menu : Survey planning fragment

Return home (click twice)



Bottom menu : exit fragment

Return to login



Encountered difficulties

- Working with hardware -> many more possible points of failure
- Connection to drone and the internet at the same time
- Switching between many different piloting interfaces (manual, automatic, emergency automatic interface)
- Remote photo and video retrieval from the drone

Possible improvements

- More/better data retrieval
- Better use of the firebase
- Better accuracy of manual piloting interface